Agenda

1. Hyperparameter tuning
2. Underflow problem
3. Feature Importance

$K = 0$ ← Laplacian Smoothing

$$P(w_i / y=1) = \frac{\text{Count}(w_i, \text{Spam}) + K}{\text{Total word in Spam} + K|V|}$$

$|V|$ → No of unique words

w_i is absent

$$P(w_i / y=1) = \frac{0 + 0}{20 + 0 \cdot |V|} = 0$$

w_i is present

$$P(w_i / y=1) = \frac{5 + 0}{20 + 0 \cdot |V|} = 0.25$$

→ Highly sensitive to training data (high variance)

↓

Overfit

$K = 10000$

w_i (word) is absent

$$\frac{\text{count}(w_i, \text{spam}) + K}{\text{Total words in Spam} + K|V|}$$

$$P(w_i / y=1) = \frac{0 + 10000}{20 + 10000 \times 100}$$

$|V| = 100$

$= 0.0099$

≈ 0.01

w_i (word) is present

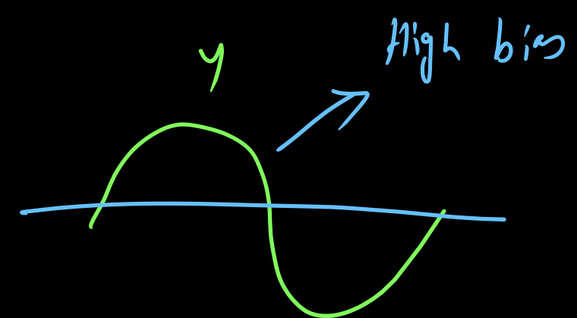
$$P(w_i / y=1) = \frac{\overset{10}{5} + 10000}{20 + 10000 \times 100}$$

$= 0.01$

→ At high K, it is INSENSITIVE to w_i (words) in training data

⇓

Underfitting



$K \downarrow\downarrow \rightarrow$ Overfitting

$K \uparrow\uparrow \rightarrow$ Underfitting

Underflow

$$P(y=1/x) > P(y=0/x)$$

$$P(x/y=1) \cdot P(y=1) > P(x/y=0) \cdot P(y=0)$$

$$P(x/y=1) = P\left(\frac{w_1}{y=1}\right) \cdot P\left(\frac{w_2}{y=1}\right) \cdots P\left(\frac{w_d}{y=1}\right)$$

$$P(x/y=0) = P\left(\frac{w_1}{y=0}\right) \cdot P\left(\frac{w_2}{y=0}\right) \cdots P\left(\frac{w_d}{y=0}\right)$$

$$0.1 \times 0.01 \times 0.2 \times 0.3 \cdots$$

$$= 0.000000142 \longrightarrow \text{Cannot it stored}$$

$$[-10^{-5}, 10^{5}] \longrightarrow \text{float}$$

$$\log\left[P(y=1/x)\right] > \log\left[P(y=0/x)\right]$$

$$\rightarrow \log\left[P(x/y=1) \cdot P(y=1)\right] > \log\left[P(x/y=0) \cdot P(y=0)\right]$$

$$\hookrightarrow \prod_{i=1}^{d} P(w_i/y=1) \qquad \prod_{i=1}^{d} P(w_i/y=0)$$

$$\log(a \cdot b) = \log(a) + \log(b)$$

$$\log P(y=1) + \log\left[\prod_{i=1}^{d} P(w_i/y=1)\right] > \log P(y=0) + \log \prod_{i=1}^{d} P(w_i/y=0)$$

$$\boxed{\log P(y=1) + \sum_{i=1}^{d} \log P(w_i/y=1) > \log P(y=0) + \sum_{i=1}^{d} P(w_i/y=0)}$$

Feature Importance

$\longrightarrow w_1 \; w_2 \; w_4$

$$P(y=1/x) \rightarrow \propto P(y=1) \cdot P\left(\frac{x}{y=1}\right)$$

$$\propto P(y=1) \cdot P\left(\frac{w_1}{y=1}\right) \cdot P\left(\frac{w_2}{y=1}\right) P\left(\frac{w_4}{y=1}\right)$$

free	money	win	hello	label (y)
0	1	1	0	1
0	0	0	1	0
1	0	1	0	1
1	0	0	1	0

$$\frac{11}{14}$$

$$\frac{P(free)}{(y=1)} \qquad P\left(\frac{money}{y=1}\right)$$

Higher Count

//

$\longrightarrow$ More frequent word, will have higher probability $\longrightarrow$ high feature Importance

Impact of Outliers

$\rightarrow$ $x_q = w_1 \; w_3 \; w_6 \; w'$ $\leftarrow$ Outlier (Rare words)

$\hookrightarrow$ laplacian Smoothing

$k = 1$

"free free free" $\rightarrow$ Spam

→ Break until 22:25PM

Present / Not Present

Multinomial Naive Bayes

$$\rightarrow P\left(\frac{w_i}{y=1}\right) = \frac{\text{Count}(w_i \text{ in } y=1)}{\text{Total word in } y=1}$$

$$P\left(\frac{\text{free}}{y=1}\right) = \frac{3}{6} = 0.5 \qquad \frac{11}{14}$$

free	money	win	hello	label (y)
~~1~~ 2/10	1	1	0	1
0	0	0	1	0
1	0	1	0	1
1	0	0	1	0

$$P\left(\frac{\text{free}}{y=1}\right) = \frac{2}{5}$$

$$P\left(\frac{\text{money}}{y=1}\right) = \frac{1}{5}$$

$$P\left(\frac{\text{win}}{y=1}\right) = 2/5$$

$$P\left(\frac{\text{hello}}{y=1}\right) = 0/5$$

Bernauli Naive Bayes

$$\rightarrow P\left(\frac{w_i}{y=1}\right) = \frac{\text{No of doc. contain } w_i \text{ in } y=1}{\text{Total no of doc in } y=1}$$

$$P\left(\frac{\text{free}}{y=1}\right) = \frac{2}{2} = 1$$

No of doc = No of rows

$$P\left(\frac{\text{free}}{y=1}\right) = \frac{2}{2} = 1$$

$$P\left(\frac{\text{money}}{y=1}\right) = \frac{1}{2}$$

$$P\left(\frac{\text{win}}{y=1}\right) = \frac{2}{2}$$

$$P\left(\frac{\text{hello}}{y=1}\right) = 0/2$$

Multinomial NB

$$\rightarrow P(w_i / y=1) = \frac{\text{Count}(w_i \text{ in } y=1) + K}{\text{Total word in } y=1 + K|V|}$$

Vocabulary

Bernauli NB

$$P(w_i / y=1) = \frac{\text{No of doc. contain } w_i \text{ in } y=1 + K}{\text{Total no of doc in } y=1 + K \cdot 2}$$

→ Classification

- ↳ Acc
- ↳ P
- ↳ R
- ↳ AUC - ROC
- ↳ PR - ROC
- ↳ F1

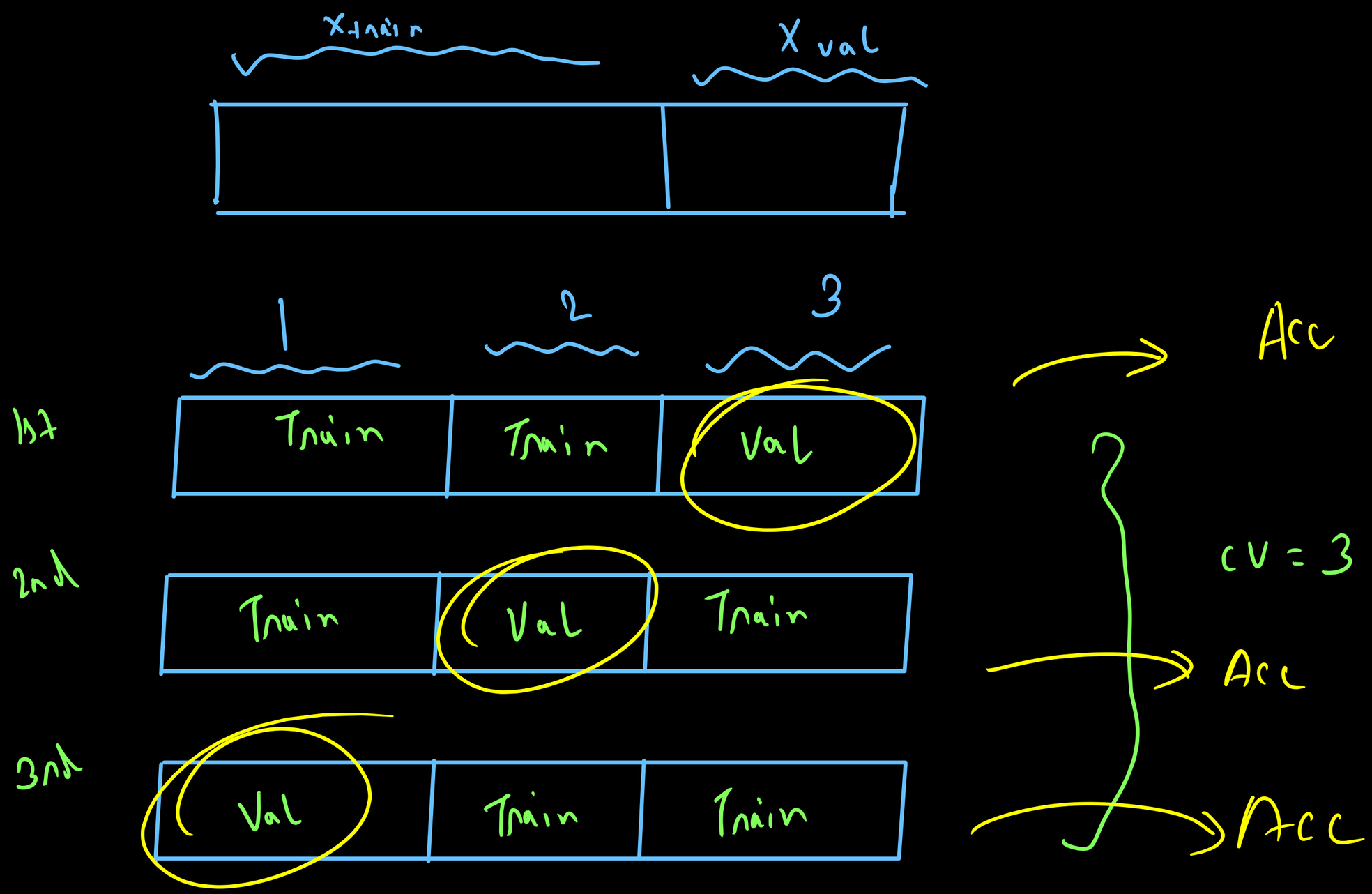
Regression

- ↳ R^2
- ↳ Adjusted R^2
- ↳ MAE/MSE

	cleaned_message
4281	u call
585	tell u headache want use 1 hour sick time
4545	never try alone take weight tear comes ur hear...
3034	raji pls favour pls convey birthday wishes nim...
2758	time iûm prob 3
...	...

	cleaned_message
2102	audrie lousy autocorrect
363	busy trying finish new year looking forward fi...
1033	oh mr sheffield wan na play game okay boss nan...
1192	night worrying appt shame missed girls night q...
2571	û_ donût worry weûll finished march û_ ish
...	...
3920	1 thing change sentence want 2 concentrate edu...

	cleaned_message
4716	k addie amp art get home
3704	ok good making money
5468	urgent last weekend draw shows å1000 cash span...
4861	free message thanks using auction subscription...
4011	discussed mother ah
...	...


X train
X val
1
2
3
1st
Train
Train
Val
Acc
2nd
Train
Val
Train
Acc
3rd
Val
Train
Train
Acc
CV = 3